ΠΥΘΙΑ · the oracle that keeps score

PYTHIA

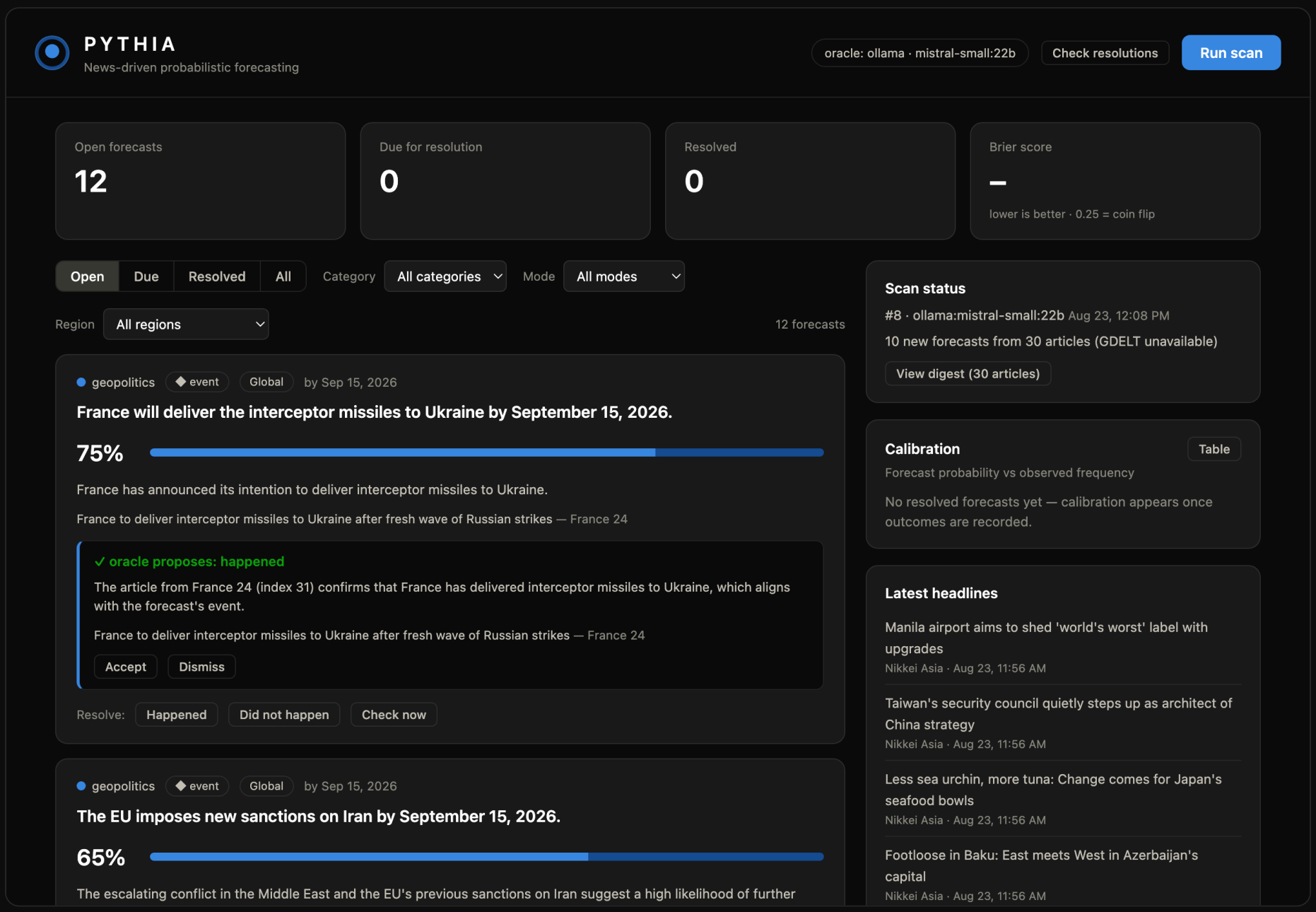
A self-measuring forecasting system. It reads the world's news, states the odds of what happens next — **and is scored on every answer it ever gives.**

Open source · runs on one laptop · swipe →

github.com/egsamaras/pythia

THE PRODUCT

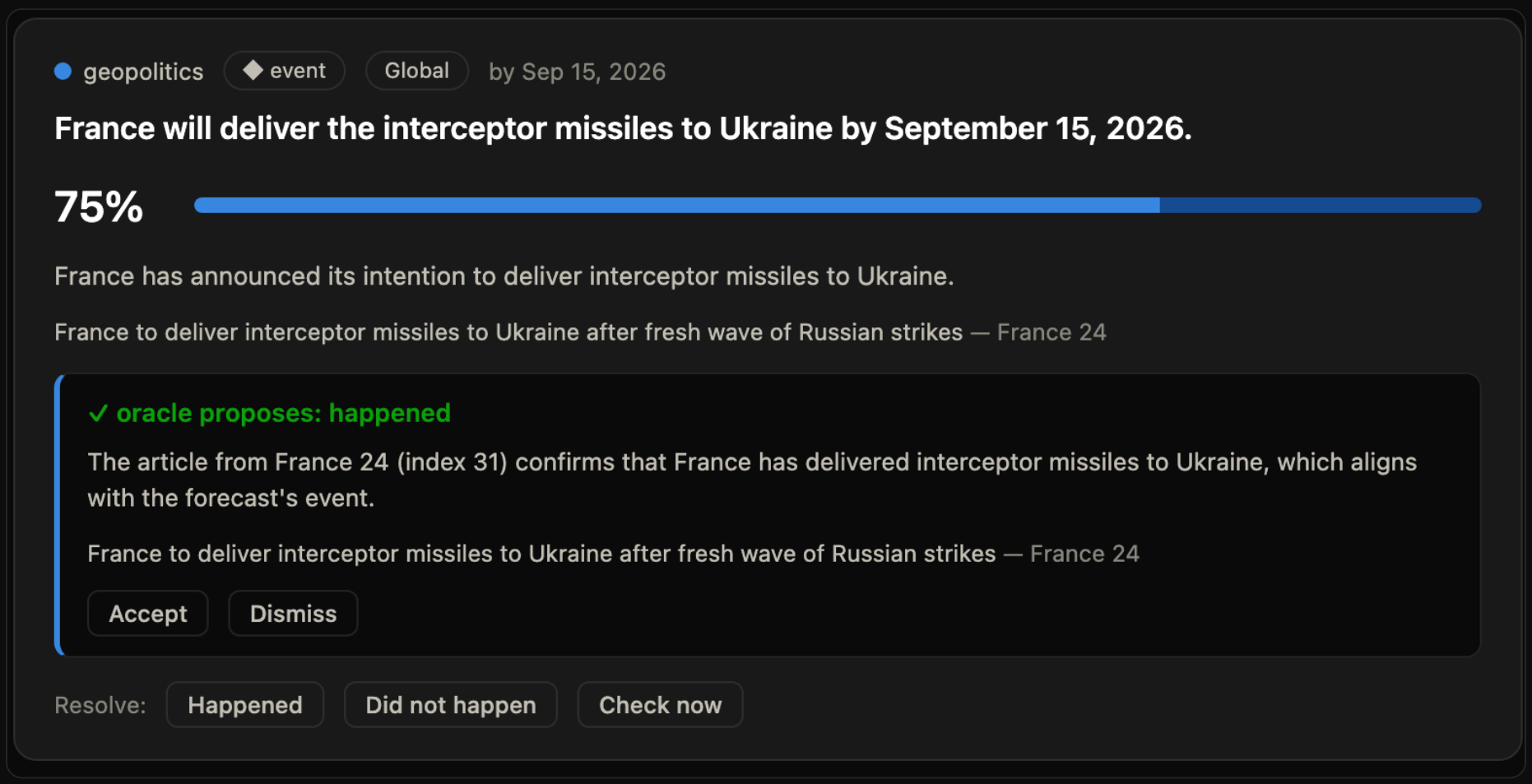
One screen, one question: can it be trusted?



Open forecasts, what's due, what's resolved — and the Brier score, empty until outcomes exist. It shows its own scoreboard whether or not the news is flattering.

ANATOMY OF A FORECAST

Every claim carries its evidence



A falsifiable statement, a probability, a deadline, the articles it reasoned from — and the oracle proposing its own resolution with a citation. You accept or dismiss; the ledger is never graded by the model being graded.

The loop

I Ingest

48 verified sources — wires, central banks, government travel advisories, conflict analysis — plus GDELT's machine-coded coverage in 100+ languages.

II Forecast

Concrete falsifiable statements with probabilities, deadlines and regions. The schema forces reasoning before the number.

III Resolve

The oracle proposes outcomes with cited evidence. A human confirms each one.

IV Update

New evidence moves open forecasts, with a full audit history and guards against drift and hindsight.

V Learn

Measured miscalibration is fed back into the forecasting prompt. It corrects itself from evidence, not vibes.

ΜΕΛΕΤΗ ΤΟ ΠΑΝ · practice is everything

Three ways it predicts

Each method is scored separately — so the ledger says not just **whether** it forecasts well, but **which kind of thinking** earns its keep.

◆ event

Will it happen?

The news says something may happen. Pythia states the odds it actually does.

→ consequence

What follows?

The downstream outcome the articles never state — with the causal chain written out and judged.

⟲ precedent

What does history say?

Reference-class forecasting: the closest past episodes, their base rate, and the adjustment for today.

I asked the model, politely and in writing, never to state a claim and its negation.

It did it anyway. **46% of my ledger** turned out to be duplicates, contradictions, or historical base rates masquerading as predictions.

Scoring both "X happens" and "X doesn't happen" guarantees one hit and one miss regardless of skill — manufacturing a respectable-looking calibration score out of pure noise.

So I stopped asking and started enforcing it in code.

ΓΝΩΘΙ ΣΑΥΤΟΝ · know thyself

What keeps it honest

Reason before number

Constrained decoding emits fields in schema order, so base rate and causal chain are written before the probability — the number conditions on the reasoning instead of justifying it afterwards.

Time-weighted scoring

A forecast is scored on its average credence across its whole open life. A last-minute jump to 95% counts only for the sliver of time it was held, so the ledger cannot be gamed.

The human gate

The oracle proposes; you resolve. Absence from a small news batch is never proof, post-horizon revisions are refused as hindsight, and every "happened" must cite its articles.

Noise never becomes instruction

Calibration feedback reaches the prompt only past 30 resolutions, and only for biases exceeding twice their binomial standard error. Below that, silence.

ΜΗΔΕΝ ΑΓΑΝ · nothing in excess

Built like it handles money

7
adversarial review rounds

148
reviewer & verifier agents

97
confirmed findings, all fixed

23
claims refuted before wasting a fix

Specialist reviewers attacked each feature — correctness, security, forecasting methodology, UX — and a second wave of agents tried to **refute every finding** before anything changed.

The survivors included a schema field-order that quietly defeated base-rate anchoring, and a scoring rule that would have let the system flatter its own record.

TAKE IT

Clone it, run it, check my work

MIT licensed. 180 tests, none of which need a network or a model. Runs entirely on a laptop against a local open model — nothing leaves the machine — with the Claude API as an optional swap.

`github.com/egsamaras/pythia`

`egsamaras.github.io/pythia`

Ask me in six months whether it can forecast.

Today all I can honestly claim is that **it will tell us.**

Forecasts are probabilistic estimates generated by a language model for research and experimentation. They are not advice — financial, investment, or otherwise. The Pythia of Delphi spoke in riddles; this one shows her working.